

# Christian Garry

Machine Learning Researcher — stochastic control · deep learning · multi-agent RL

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## Profile

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MSc Scientific Computing researcher (Distinction track, ~86%) working at the intersection of **deep learning**, **stochastic control**, and **quantitative finance**. My dissertation builds and debugs **deep neural solvers for a competitive market-making game**, validates them to **0.135%** against an exact benchmark, and uses **multi-agent reinforcement learning** to study how trading algorithms learn to price — finding a reproducible, statistically-significant *supra-cartel* effect. I move fluently between **model architecture**, **feature/transformation design**, and **stochastic-control theory**, and I hold the work to a high evidentiary bar: ablation batteries, multiple-comparison-corrected statistics, and a machine-checked error theory. Strong communicator — a 10-chapter dissertation plus three standalone papers.

## Research

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**Deep BSDEs for Mean-Field Market Making — MSc Dissertation, Durham University · 2026**

*Stochastic control × deep learning × multi-agent RL · [github.com/cgarryZA/MFG-BSDE-Equilibrium](https://github.com/cgarryZA/MFG-BSDE-Equilibrium)*

- Built **deep backward-SDE-with-jumps solvers** (PyTorch) for the Cont-Xiong dealer market-making game — pricing/quoting under inventory risk and mean-field competition — matching the exact tabular solution to **0.135% spread / 0.216% value error** and **scaling to ~5,000 agents** where grid methods are infeasible.
- **Tinkered architectures into working models**: identified, characterised (two as propositions) and fixed **three independent failure modes** that silently kill the mean-field signal in naïve networks — *generator bypass* (gradient path severed), *BatchNorm erasure* (zero-variance broadcast features), and *DeepSets collapse* (symmetric inputs cancelling at  $O(1/N)$ ). Post-fix, the learned competition factor varies  $4\times$  and optimal quotes shift  $2\times$ ; validated by a **26-experiment ablation battery**.
- **Trading-strategy learning dynamics**: trained **20 seeded decentralised-MADDPG** agents on the same game → mean spread **1.866** (95% CI [1.654, 2.072]), above both Nash (1.515) and cartel (1.593); **t=3.22, p=0.0022**, Cohen's d=0.72, surviving Bonferroni & Holm. The game has a unique competitive equilibrium with no supra-cartel fixed point — so the effect is a property of the **learning algorithm**, not the market (direct algorithmic-collusion / surveillance relevance).
- Developed an **a-posteriori error-certification theory** for deep-BSDE solvers — a computable residual estimator two-sidedly equivalent to the true error, with no spurious minima and a certified training-stop rule — and **machine-checked its core in Lean 4 / Mathlib** (~36k LOC, axiom-clean; 4-wave adversarial audit caught a real soundness bug).
- Built the **reproducible-research infrastructure**: a CI-guarded data→paper pipeline (build fails if any cited number drifts from its source), a GPU overnight job queue, and remote automated job submission.

**Silicon Carbide JFET CPU — MEng Dissertation, Durham University · 2023–24 · 84%**

- Designed and simulated a 4-bit CPU in SiC JFET logic and built the **full verification toolchain** (C++ compiler, assembler, Python automation, emulator parity + waveform-to-state reconstruction) — instruction-level validation against a behavioural reference. Engineering rigour from device models to verified system behaviour.

## Education

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**MSc Scientific Computing & Data Analysis (AI for Engineering) — Durham University · Sep 2025 – Sep 2026**

- Tracking **Distinction** (~86% across completed modules). Standout marks: **95%** Performance Modelling/Vectorisation/GPU, **94%** Machine Learning & Statistics, **~89%** in both Advanced Bayesian ML modules.
- Coverage directly relevant to ML research on large datasets: **Bayesian inference** (MCMC, variational methods, Gaussian processes), **deep learning** (CNNs, Transformers, diffusion, LoRA), **convex optimisation** (duality, KKT) applied to estimation & SVMs, **random forests/boosting**, and **GPU/HPC performance engineering** (roofline, AVX-512, CUDA, LIKWID).
- Python, C, R; reproducible, tested workflows.

**Mathematics modules (NetMath) — University of Illinois Urbana-Champaign · 2026**

- Credit-bearing, exam-assessed proof-based mathematics taken alongside the MSc: **MATH 314** Introduction to Higher Mathematics (in progress, current average 100%) and **MATH 447** Real Analysis (upcoming) — sequences and convergence, metric spaces, compactness, and Riemann integration. Deepens the analytic foundations under the stochastic-control and ML theory.

**MEng Electronic Engineering — Durham University · 2020–2024 · Upper Second-Class Honours (2:1)**

## Experience

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**Graduate Communications Engineer — Siemens PLC · Sep 2024 – Present**

- Designed and built a **Retrieval-Augmented Generation platform from scratch in Python** (local LLMs via Ollama + Qdrant vector search) over large protocol specs and source trees — **cutting engineer time-to-answer by >90%**, with an emphasis on transparent, reproducible behaviour.
- Implemented **C/C++ communication components** (TCP/IP, serial) for digital-twin simulation, streaming data between simulated systems and PC tooling.

## Technical Skills

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**ML / modelling:** deep learning (PyTorch, FBSNN/deep-BSDE, CNNs/Transformers), architecture & feature debugging, multi-agent RL (MADDPG), Bayesian inference (MCMC, Gaussian processes), random forests/boosting, hypothesis testing with multiple-comparison correction, bootstrap CIs.

**Stochastic / quant:** BSDEs/BSDEJs, McKean–Vlasov & mean-field games, HJB↔BSDE stochastic control, Monte-Carlo, market microstructure / optimal market making, inventory risk.

**Engineering:** Python, C++, C, R; GPU/CUDA, HPC (roofline, vectorisation, Slurm); Git/CI, reproducible-research pipelines; Lean 4 / Mathlib (formal verification).